

#7-R: Pathway-aware HAN for Alopecia Metabolite Discovery

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Background & Motivation

Alopecia

- Hair loss concerns of millions
- Existing treatments have limited coverage of potential candidates

Yeast

- Saccharomyces cerevisiae
- Provide 1000+ well-annotated metabolites with rich pathway data

The Gap

- Yeast metabolite ↔ Alopecia target links are unavailable experimentally
- Ambiguous & Large data for rank

Our new approach

We propose a **Heterogeneous Attention Network (HAN)** with **Random Walk Positional Encoding (RWPE)** to bridge yeast metabolites and alopecia drug-target network. And training is based on **Tanimoto score (Structural similarity)**.

HAN

+

Tanimoto Score

+

RWPE

→

Pathway-level signal discovery

HAN + RWPE Structure Model

Heterogeneous Graph Structure

Metabolite

S.cerevisiae

Tanimoto →

Drug

Known treatments

DB →

Target

AR, SRD5 ...

DB →

Disease

Alopecia

- GNN learns node embeddings via HAN (Heterogeneous Attention Network)
- Dot product of embeddings = Relevance Score

Random Walk Positional Encoding

$T = D^{-1}A$

→

$RWPE[i,k] = (T^k)[i,i]$

→

2048 bits

Morgan FP

40 bits

RWPE

Adjacent transition prob. Prob of returning node i after k steps Concat with Structure Feature → Final Embedding

RWPE[i,k] encodes how likely a random walk starting from metabolite i returns to itself after k steps capturing its structural position within the pathway graph

Model Training

Training Loss

Validation AP score

Best Val AP : 0.962 , Metabolite → Drug AUC : 0.931, Drug → Target : 0.992

Training loss converges & Validation Scores are good within 30 epochs.

→ Model successfully learns both Metabolite-Drug (Tanimoto-based) and Drug-Target (DB-based) relationships

Tanimoto Score & Metabolite Ranking

Candidate selection

All yeast metabolites 1000+

Background 368 (Valid Metabolites)

Foreground 100 (Analysis Target)

ATP Acetyl-coA → Remove currency / hub / artifact GNN top ranked

Tanimoto Score Distribution

GNN score

Metabolite	GNN rank
myo-inositol	1
zymosterol	13
lanosterol	17
adenosine	38
spermidine	87

Pathway Enrichment

Drug evidence score

Metabolite	Drug rank	Evidence
adenosine	#1	Alopecia clinical trial
spermidine	#2	Hair growth factor (Anagen)
lanosterol	low	Weak direct evidence
Myo-inositol	low	Weak direct evidence

Top 7 enriched Pathway

Pathway Enrichment

Null model comparison : Sum of Top 10 pathway's -log(p)

Actual GNN : 30.68

Null distribution : 5~18

Random Null
Randomly sample 100 metabolites from background Repeat 1000 times → Build null distribution

Degree-matched Null
Sample 100 metabolites with same pathway-degree distribution as GNN top 100 → Controls for degree

Parameter Tuning

Metabolite	dim=20	dim=40	dim=50	dim=100	Consistency	Relativity with Alopecia
RWPE enhanced						
spermine	—	+97	+90	—	2/4	High (anagen elongation)
spermidine	—	—	+87	—	1/4	High (anagen elongation)
O-succinyl-L-homoserine	+149	+109	+137	—	3/4	Low (methionine)
gamma-L-glutamyl 5-phosphate	+176	+136	—	—	2/4	Low (glutamate)
(L-cysteinyl)adenylate	—	—	+93	+146	2/4	Moderate (cysteine/keratin)
(L-seryl)adenylate	—	—	+91	+143	2/4	Low (serine)
RWPE suppressed						
(R)-acetoin / (S)-acetoin	-156	-198	-191	-139	4/4	None (butanediol pathway)
(R)-lactate / (S)-lactate	—	-176 / -170	-193 / -177	-139 / -130	3/4	None (general metabolite)
isobutanol	-129	-164	-167	—	3/4	None (artifact)
aci-nitroethane	-112	-153	-160	-148	4/4	None (artifact)
sn-glycero-3-phosphocholine	—	—	—	-176	1/4	High (phospholipid)

Validation AP

RWPE Dim=20 Val AP = 0.896

Validation AP

RWPE Dim=40 Val AP = 0.906

Validation AP

RWPE Dim=100 Val AP = 0.890

Conclusion & Future Work

Classical Approach:

Tanimoto (Only measure structure similarity)

Our approach:

2048 bits

Morgan FP

40 bits

RWPE

→

HAN

Result

- HAN learned the Tanimoto algorithm accurately (Best Val AP: 0.962, Metabolite-Drug AUC: 0.931).
- HANConv is non-convex optimizing algorithm (Not optimal, sensitive to initialization)
- HAN reflected the biological pathway as we expected through Pathway Enrichment (Actual GNN score: 30.68 vs. Null: 5~18).
- We devise new approach: RWPE + HAN
- RWPE embedding incorporated new structural information, capturing specific reactions of the pathway (Peak Val AP: 0.906 at Dim=40, dUTP detection).

Future Work

Robustness Validation

RWPE Dimensionality Analysis

Biological Verification